Boddu Hadassah

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EDUCATION

Institution	Degree / Board	Passing Year	Score / CGPA
Indian Institute of Technology Hyderabad	B.Tech	2027	7.53
Narayana Junior College, Vijayawada	Class XII (Board of Intermediate Education AP))	2023	96.10%
Narayana High School, Vijayawada	Class X (Board of Secondary Education AP)	2021	100.00%

WORK EXPERIENCE

AI Engineer Intern | *ORBCOMM*

Hapag POC

- Analyzed **~1M container assets** across major US ports to derive dwell times, container movement patterns, and stoppage locations for supply-chain monitoring.

- Built scalable **data processing** pipelines using **PySpark** and Microsoft Fabric, including telemetry cleaning, validation, Delta Lake ingestion, and **geocoding** of **~718K** location records.

- Engineered a parallelized **stoppage-detection** algorithm using PySpark window functions, reducing processing time from hours to minutes and enabling analysis across **1M+ assets**.

Orbcomm Geofence Library (OGL)

- Independently designed and built an **end-to-end** geofence management platform to replace error-prone manual database operations, enabling teams to create, edit, visualize, version, and validate geofences through a centralized application.

- **Key features:** Interactive map-based geofence drawing and editing, zero-lag caching, role-based access control, automated overlap detection, and version history with one-click rollback.

SKILLS

Languages: C, C++, Python, SQL

Libraries: Numpy, Pandas, Matplotlib, OpenCV.

Frameworks: Microsoft Fabric, PySpark, Django, Tensorflow, keras and scikit-learn,

Machine Learning: Supervised Learning, Neural Networks, Feature engineering, Transformers.

Soft Skills : Open-minded and have collaborative approach to work with diverse groups

Positive attitude towards learning, with clarity in thoughts and effective communication.

PROJECTS

Vehicle rental system

- Developing a **Full-stack Rental system** using Django, HTML/CSS, and SQL for seamless vehicle management and customer booking workflows.

- Dynamic vehicle booking and availability tracking features with **real-time** total-cost computation and SQL validation.

- Optimized **database schema** and relationships to ensure data integrity and efficient query performance.

GitHub Repo Analyser

- Designed an **AI agent** using **LangChain** to analyse GitHub repositories with **RAG-based retrieval**

- Enabled the agent to generate README files and answer complex codebase queries for effortless repo exploration.

- Built a **FastAPI** backend and **deployed** it on **AWS** for reliable and scalable access.

Machine Learning Classification of CHIME Fast Radio Bursts

- Performed **feature extraction** and **Data Analysis** on CHIME fast radio burst (FRB) data.

- Applied Logistic Regression, SVM, XGBoost and AdaBoost to classify FRBs as repeating or non-repeating.

- Achieved around **0.92 - 0.96 accuracy** on multiple algorithms.

Real-Time Facial Emotion Recognition

- Developed a real-time facial emotion recognition system using **CNN** on the **FER-2013** dataset, achieving 70% accuracy

across 7 emotion categories.

- Integrated face detection with **OpenCV's** Haarcascade and **live webcam** feed for emotion classification.

- Trained a multi-layer CNN model and Applied early stopping and model checkpointing to optimize performance.

SCHOLASTIC ACHIEVEMENTS

- Achieved **13th rank** with **Distinction** at the **zonal level** in the SOF NSO Olympiad and awarded a **Gold Medal**.

- Awarded a **Bronze Medal** in the SOF IMO Olympiad.

- Qualified for **AMTI** level 2 (Among **top 1%** from the level 1).

- Cleared **National-Level** Little Teachers Competition and recognized among School's **Best Speakers**.

- Consistent academic excellence (Grades 1-10) with top JEE Mains performance.

POSITIONS OF RESPONSIBILITY

Student Representative | NSS IITH -**Headed NSS** and organized various social service initiatives and events.

Design & Multimedia Head | Alumni Cell IITH - Led the **design team** and managed creative deliverables.

UG Mentor | Sunshine IITH - Mentored and guided **15+ junior students**.

Events and Outreach Core | Cepheid IITH - Conducted and promoted **space-related events and activities**.

EXTRACURRICULAR

- **NSS Head, IITH:** Leading and coordinating **1,600+ NSS volunteers** and organizing events.

- Swimming: IITH Swim League **Winner**; skilled in **deep-water** swimming **(14 ft)**.

- Arts & Wall Painting: Skilled in wall painting, sketching, and creative arts.